

ServiceNow Certified Application Developer

SmartInternz

Project Title

**Automated Network Request
Management in ServiceNow**

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Project overview

The **Automated Network Request Management in ServiceNow project** focuses on simplifying and accelerating the handling of network-related service requests. By leveraging ServiceNow's service catalog, workflows, and automation capabilities, the solution reduces manual effort and ensures that requests are processed efficiently. Users can submit network-related requests through a self-service interface, while automated routing, approvals, task assignments, and notifications streamline the fulfillment process. The system improves request tracking, reduces processing delays, and provides better visibility for both users and IT teams.

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1. Introduction

Modern organizations rely heavily on efficient network services to support their day-to-day operations. As businesses grow, the number of network-related service requests, such as access requests, configuration changes, connectivity issues, and infrastructure support, also increases. Managing these requests manually can lead to delays, errors, repeated follow-ups, and limited visibility into the request lifecycle.

To address these challenges, the Automated Network Request Management in ServiceNow project provides a streamlined solution for managing network-related service requests. The system uses ServiceNow's service catalog, workflow

automation, approval processes, task assignments, and notifications to simplify request handling.

Users can submit requests through a self-service interface, while the system automatically routes requests to the appropriate teams, manages approvals, assigns tasks, and provides status updates. This reduces manual effort, improves processing accuracy, and provides better transparency for both requesters and IT teams.

The project demonstrates how ServiceNow can be used to automate and manage IT service processes in a structured and efficient manner.

2. Project Objectives

The main objective of the Automated Network Request Management in ServiceNow project is to develop a structured and automated system for managing network-related service requests. The project aims to reduce manual effort, improve request processing, and provide better visibility throughout the request lifecycle.

The system allows users to submit network service requests through a self-service interface. Based on the request details, the system automatically routes the request to the appropriate team, initiates approval processes, assigns tasks, and sends notifications to the concerned users.

The key objectives of the project are:

- To automate the submission and processing of network service requests.
- To reduce manual effort and minimize errors in request handling.
- To implement automated approval and task assignment processes.
- To provide notifications and status updates to users.
- To improve transparency and tracking of requests.
- To demonstrate the use of ServiceNow workflows, service catalog, and automation features.

3. Key Features

- Custom service catalog for network-related requests

- User-friendly forms for capturing request details
- Automated approval workflow for network requests
- Automatic task assignment to the appropriate team
- Email notifications for request updates and approvals
- Real-time tracking of request status
- ServiceNow Flow Designer for process automation
- Improved transparency and reduced manual effort

4. ServiceNow Developer Setup:

Access the ServiceNow Developer Portal

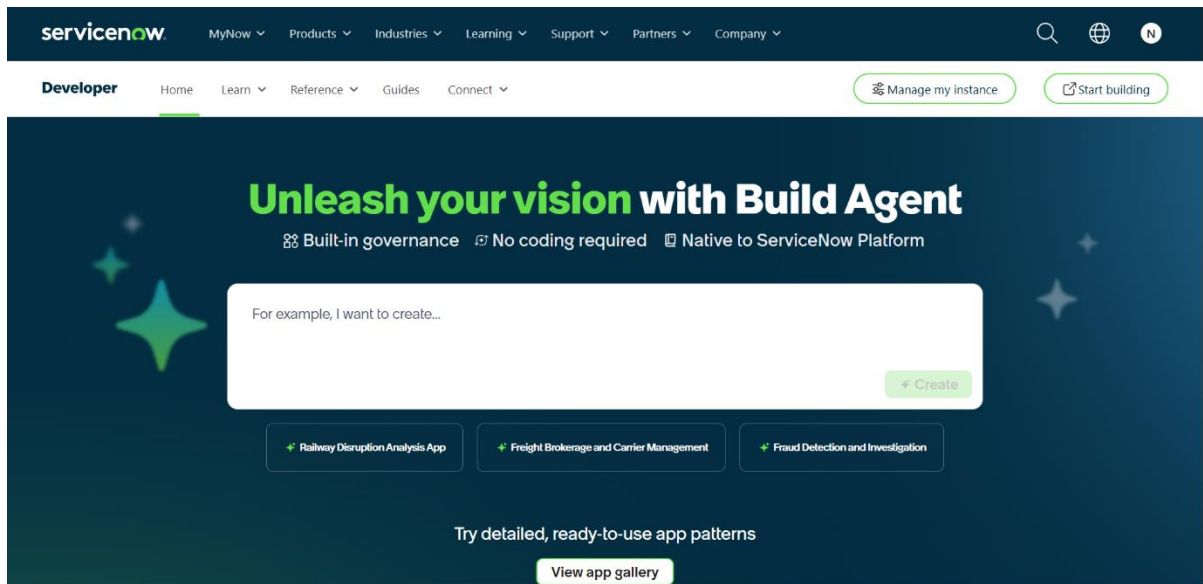
1. Go to ServiceNow Developer Portal(<https://developer.servicenow.com/dev.do>).

Sign up for a free developer account and fill out the following details.

2. After signing up, you will get a verification mail to your provided email id. After the verification your ServiceNow Developer Portal Home Page will appear.

3. Now click on start building it will take you to the section where you can request a Personal Developer Instance (PDI) or start using App Engine Studio and other tools.

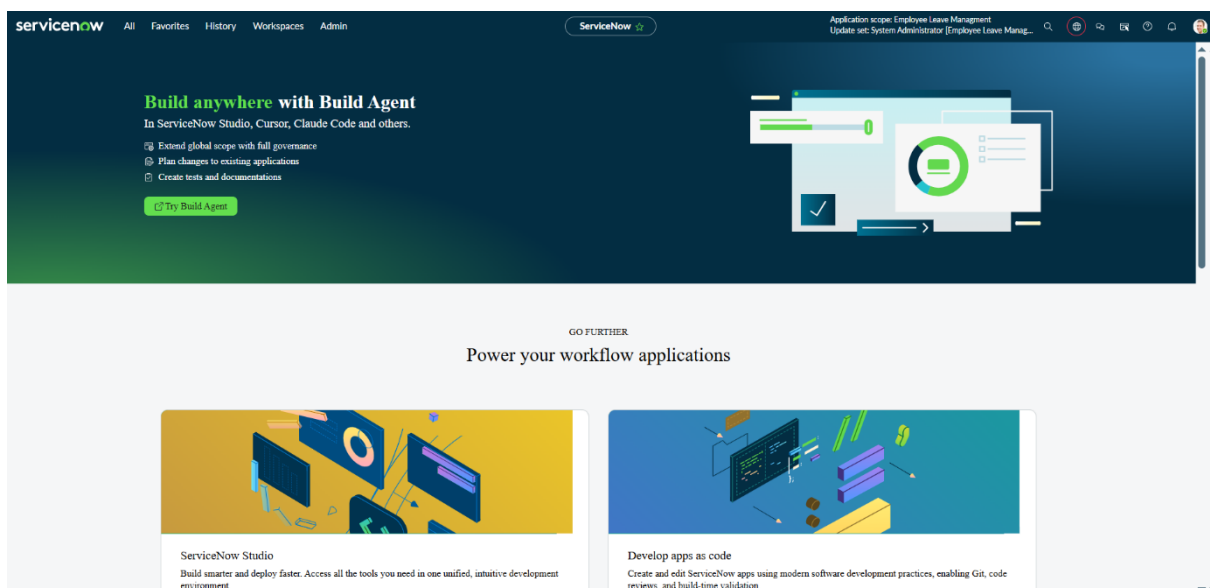
4. **Profile Icon(Top Right Corner)** → Manage your account, request instances, and check your developer profile.



5. Project Implementation in ServiceNow:

After the instance building is completed, the page will be directed into your Creator Studio.

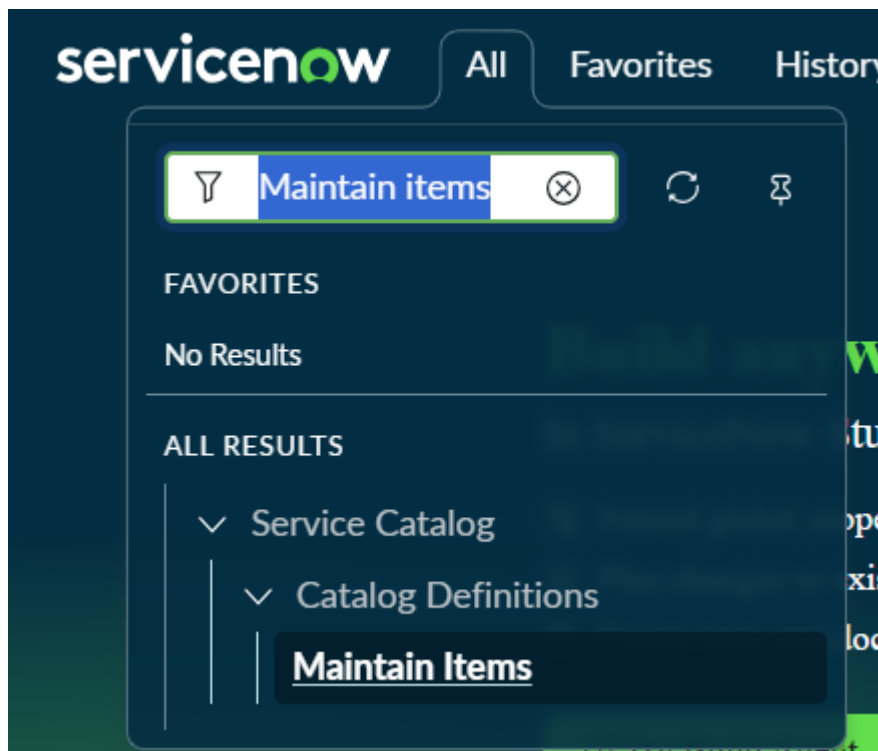
Creator Studio in ServiceNow provides a **guided, no-code environment** to build applications quickly. It is especially used for creating **request-based applications** by defining forms, setting up tables, and automating workflows.



a. Service Catalog Creation

i: Catalog Creation

1. Navigate to Application navigator
2. Click on All >> search for Service Catalog
3. Under Service Catalog >> Maintain items
4. Click on New
5. Fill the details >> Name – Network Request
6. Select Catalog >> Service Catalog
7. Select Category >> Network Standard Changes
8. Fill the Short Description as Network Request Management.
9. Click on Save.



Click on New

Name	Short description	Active	Roles	Catalogs	Category	Price	Type	Updated
Network Request	Network request Management	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-08-11 01:40:14
Reboot Windows Server	Reboot a Windows Server (after patching ...)	true		Service Catalog	Server Standard Changes	\$0.00	Item	2026-07-27 22:24:49
Decommission local office Domain Controller	Decommission a server from use including...	true		Service Catalog	Server Standard Changes	\$0.00	Item	2026-07-27 22:24:49
Change VLAN on a Cisco switchport	Change the port of a Cisco switch to a n...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-07-27 22:24:49
Clear BGP sessions on a Cisco router	Resend the complete BGP table to neighbo...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-07-27 22:24:49
Add network switch to datacenter cabinet	This standard change template describes ...	true		Service Catalog	Network Standard Changes	\$0.00	Item	2026-07-27 22:24:48
Manage Knowledge Ownership Group	Catalog item for managing knowledge owne...	true			(empty)	\$0.00	Item	2026-02-02 20:46:24
Create a Standalone Schema	Create an Avro Schema on the instance ba...	true		Service Catalog	(empty)	\$0.00	Item	2026-02-02 20:46:17
Create a new Export Set	Create a new Export Set to export data f...	true		Service Catalog	(empty)	\$0.00	Item	2026-02-02 20:46:17
Register a Business Application	Register a new business application into...	true		Service Catalog	Business Application Lifecycle Management	\$0.00	Item	2026-02-02 20:46:16
New LDAP Server	Create a new LDAP server record	true		Service Catalog	(empty)	\$0.00	Item	2026-02-02 20:46:16
Catalog UI Policy Actions	Catalog UI Policy Action step inside Cre...	true			(empty)	£0.00	Item	2025-09-29 02:17:27
Composite record producer to	A composite record producer to							

Create the Network Request

Build and modify items faster with the improved Catalog Builder

Catalog items are goods or services available to order from the service catalog. Items can be anything from hardware, like tablets and phones, to software applications, to furniture and office supplies.

- Enter a Name and Short description to display for the item.
- Enter a Price, approvals, variables, and other information as needed.

Name: Network Request

Catalogs: Service Catalog

Category: Network Standard Changes

State: None

Checked out: None

Owner: System Administrator

Application: Employee Leave Management

Active: ☒

Fulfillment automation level: Unspecified

Item Details: Process Engine | Picture | Pricing | Portal Settings

Short description: Network request Management

Description:
 B I U
 Font: Verdana, 14pt

ii: Variables Configuration

Open the catalog item just created.

Scroll down to the **Variables** related list and click **New** to create form fields.

1. Select Variables type as Single line text, Multi line text, Reference, Container Start etc as per requirement
2. Catalog Item – Network Request
3. Order – 100,200,300,,,
4. Question – Provide the variable label

- Follow the same procedure as we used for Variables Creation, for the variable set as well.
- Apply variable sets to the catalog item.

Variables (10)	Variable Sets (1)	Catalog UI Policies (2)	Catalog Client Scripts	Available For	Not Available For	Categories (1)	Catalogs (1)	Catalog Data Lookup Definitions	Related Articles	Related Catalog Items	Assigned Topics
Order ▾ Search Actions on selected rows... New Edit											
Catalog Item = Network Request											
Variable set Order ▴ Requester information 100 1 to 1 of 1											

Variables Types

1. Opened on behalf of >> Reference>> reference to user table
2. Email Id >> Single line text >> Auto populate by Opened on behalf of variable.
3. User name >>Single line text >> Auto populate by Opened on behalf of variable.
4. Phone Number >>Single line text >> Auto populate by Opened on behalf of variable.
5. Proof of Document >> Attachment

Variables (5)	Catalog UI Policies	Catalog Client Scripts	Included In (1)	Catalog Data Lookup Definitions
Order ▾ Search Actions on selected rows... New				
Variable set = Requester information				
☐	Name	Type	Question	Order ▴
	user_name	Single Line Text	User name	
	opened_on_behalf_of	Reference	Opened on behalf of	
	phone_number	Single Line Text	Phone Number	
	email_id	Single Line Text	Email Id	
	proof_of_document	Attachment	Proof of Document	
1 to 5 of 5				

V: Catalog UI Policy Configuration

Scenario: If user selects types of devices is **Others**, then Please specify field should populate.

Procedure:

1. Navigate to catalog items
2. Open Network Request item

3. In related list, we have Catalog UI policy
4. Click on New button to configure New UI policy
5. Select Applies to as Catalog item
6. Select catalog item as Network Request
7. Provide short description, if required
8. Apply condition>> **types of devices** is **others**
9. Click on save, after saving the form will get UI policy actions in the related list
10. Click on New button to configure new UI Policy action, and Select the variable which we want to display on condition
11. Make Visible True as per our requirement
12. Update the UI Policy and Test the same on Catalog form

Short description	Variable set	Conditions	Reverse if false	On load	Inherit	Updated	Order
Types of devices is others	(empty)		true	true	false	2026-08-11 01:38:53	100
Relocation fields hiding	(empty)		true	true	false	2026-08-11 01:40:14	100

b. Creation of Table

i. Creation of Table

1.Navigate to: System Definition > Tables.

a.Click New to create a new table.

b.Fill in Table Information:

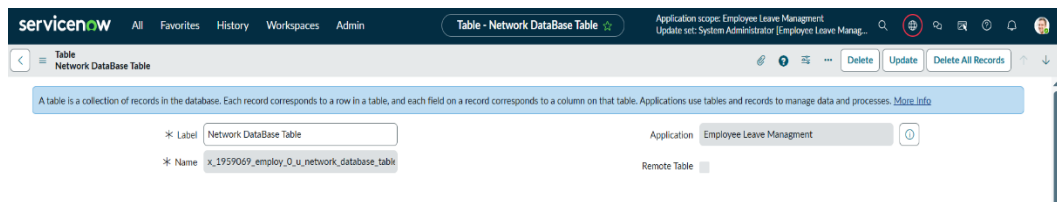
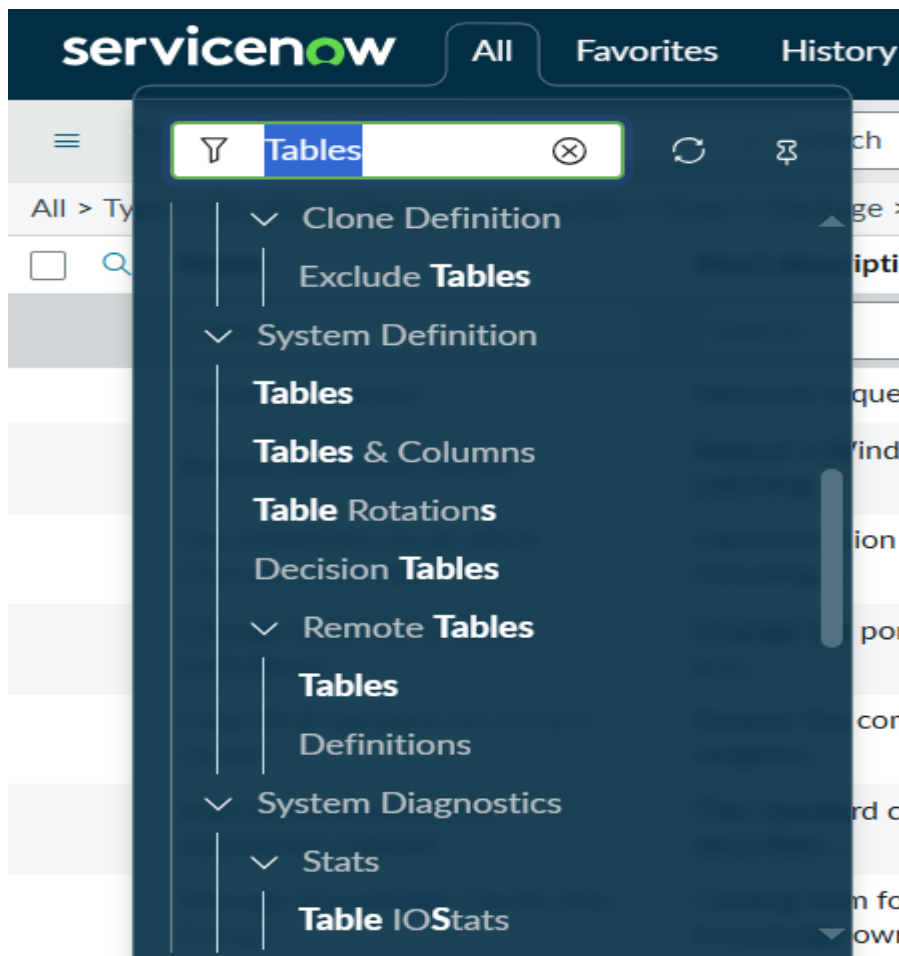
i.Name: Name of the table -----

ii.Label: Backend name of the table-----

iii.Auto-generate schema: Leave it checked if you'd like ServiceNow to

auto-generate schema fields.

c.Click Submit to create the table.



ii. Creation of fields

In ServiceNow, fields are created at the **table** level. To create a field, you first need to identify the table where the field will reside.

1. In the **Application Navigator** (left-side panel), type **Tables** in the search bar.
2. Under **System Definition**, click **Tables**. This will take you to a list of all tables in the system.

a. Select the Table to Add the Field

From the list of tables, search for and select the **table** you want to add a field to. For example, if you want to add a field to the **Network database** table:

1. Type "**Network database**" in the search box or scroll through the list.
2. Click on **the Network database** table name. You'll now see a list of all fields (columns) associated with the **Network database** table.

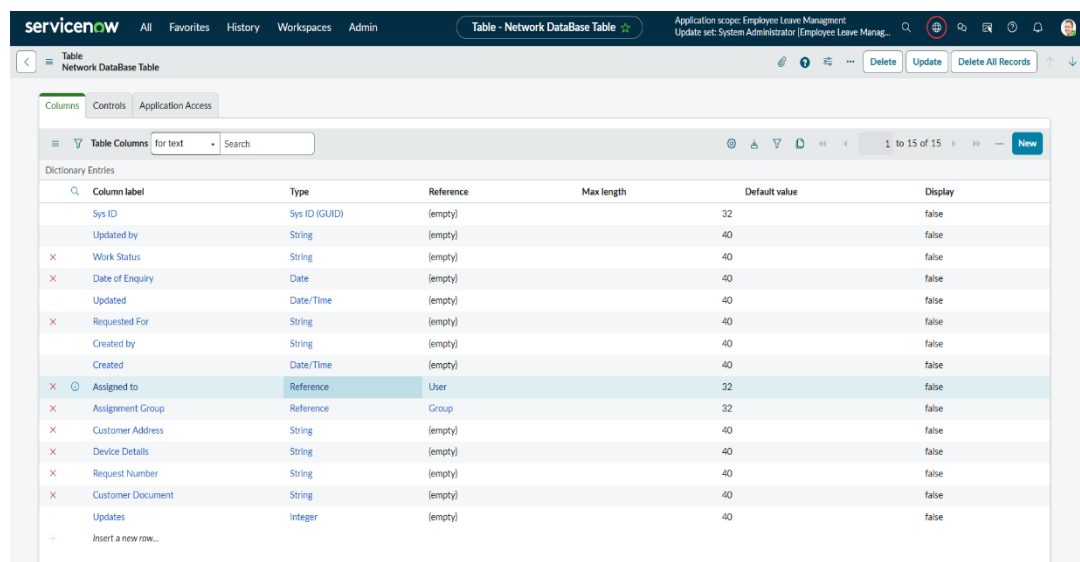
b. Open the Table's Columns

After selecting the table, you'll be brought to a view that lists all the columns (fields) that currently exist on that table.

To create a new field (column), go to the **Columns** tab (this is where all fields for the selected table are listed).

c. Create a New Field

1. In the Columns tab, click the New button located at the top-right corner of the page to create a new field.
2. You'll now be prompted with a form where you need to define the new field. The following fields need to be filled out:

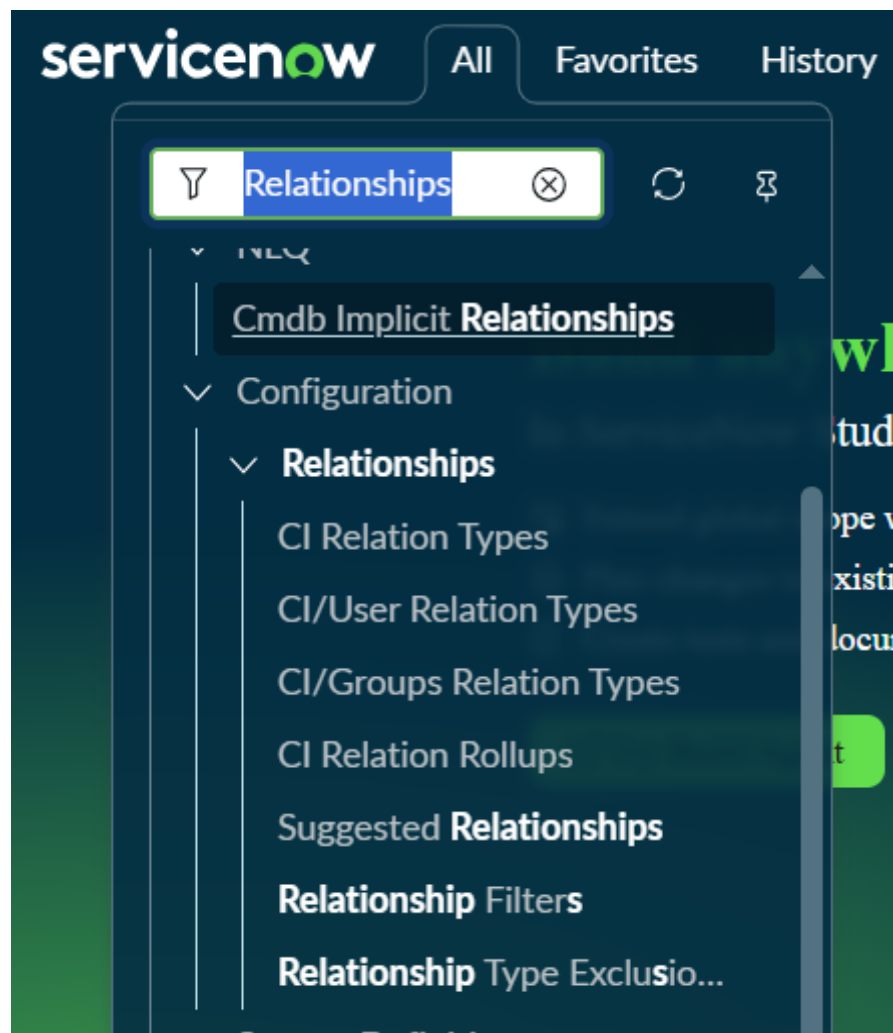


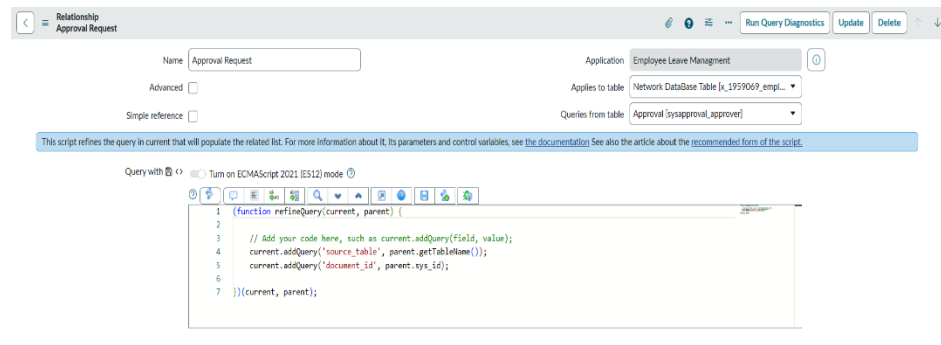
Column label	Type	Reference	Max length	Default value	Display
Sys ID	Sys ID (GUID)	(empty)	32		false
Updated by	String	(empty)	40		false
Work Status	String	(empty)	40		false
Date of Enquiry	Date	(empty)	40		false
Updated	Date/Time	(empty)	40		false
Requested For	String	(empty)	40		false
Created by	String	(empty)	40		false
Created	Date/Time	(empty)	40		false
Assigned to	Reference	User	32		false
Assignment Group	Reference	Group	32		false
Customer Address	String	(empty)	40		false
Device Details	String	(empty)	40		false
Request Number	String	(empty)	40		false
Customer Document	String	(empty)	40		false
Updates	Integer	(empty)	40		false

c. Request Approvals Creation(Related List)

i. Creation of Related List

1. Navigate to **System Definition > Relationships**.
2. Click **New** to create a new relationship.
3. Fill in the following details:
 - o **Name**: Approval Request
 - o **Applies to Table** : Network Database table.
 - o **Queries from Table** : Sysapprovals table.
 - o **Active**: Make sure it's set to True.
3. Save the relationship.





ii. Adding Related List to the Table

1. Navigate to **Form Designer** for the table where you want to show related records.
2. Add a **Related List** widget to the form.
3. Select the **Related List** you want to show.

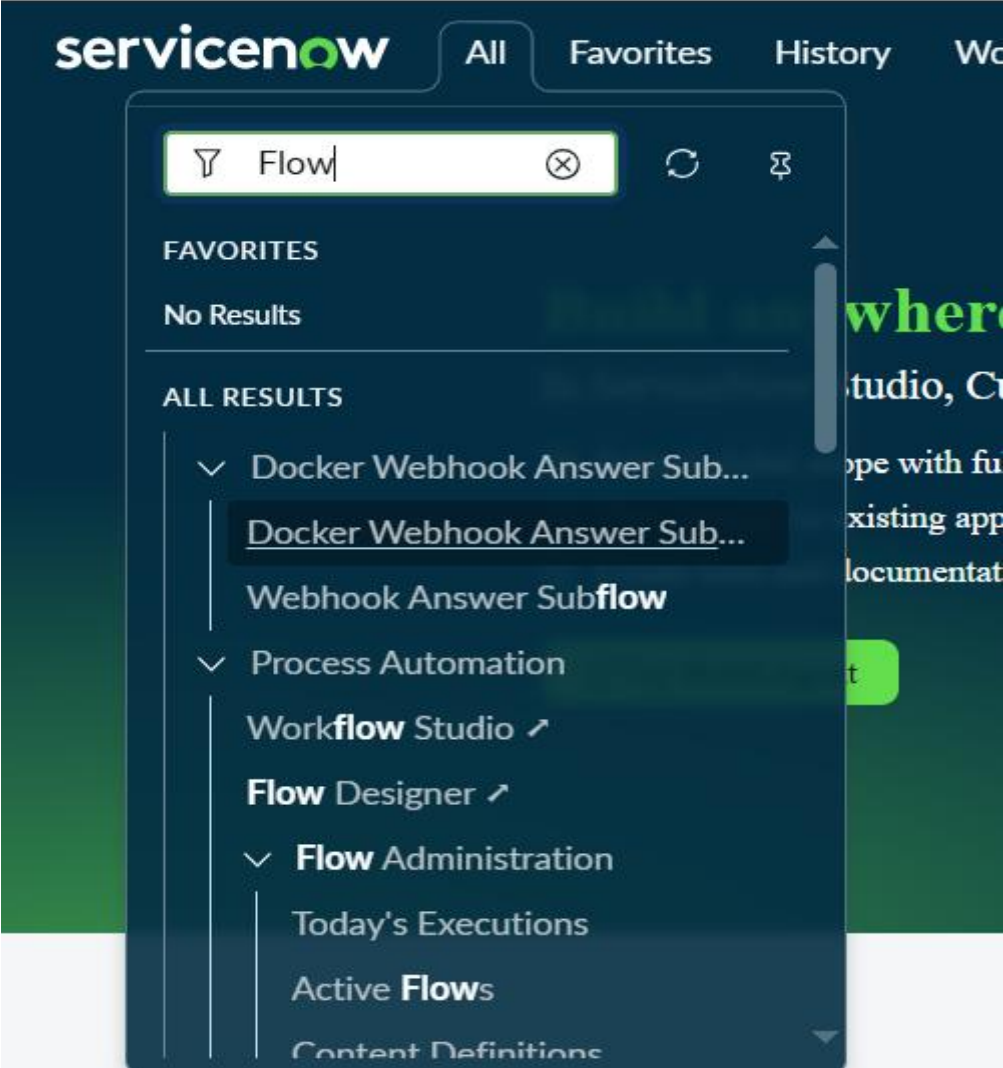
d. Creation & Implementation of flows, Actions in Flow Designer

i. Creation of Flow

1. Navigate to Flow designer home page
2. Click on New to create a new flow
3. Provide flow name as **Network Request**
4. Provide description of flow
5. Click on Build flow.

ii. Configuring Trigger

1. Click on (+) Icon to Configure the Trigger
2. Select Trigger as Application >> Service catalog
3. Click on **Done**.



Workflow Studio

Homepage Operations Integrations

Playbooks Flows Subflows Triggers Actions Decision tables

Flows 96
Last refreshed 2m ago.

Name	Application	Status	Active	Updated	Updated by
Application Intake Request Flow (Deprecated)	Application Intake	Published	false	2026-07-27 22:45:49	system
Application Intake Request V2	Application Intake	Published	true	2026-07-27 22:45:42	system
Benchmark Recommendation Evaluator	Benchmarks Spoke	Published	true	2026-02-02 20:07:22	system
Business process approval flow	Global	Published	true	2020-09-27 22:06:13	admin
Change - Cloud Infrastructure - Authorize	Global	Published	true	2020-11-11 07:08:05	admin
Change - Conflict Detection	Global	Published	true	2026-02-02 19:59:23	system
Change - Emergency - Authorize	Global	Published	true	2020-10-06 05:39:49	admin
Change - Emergency - Implement	Global	Published	true	2020-09-23 05:06:26	admin
Change - Emergency - Review	Global	Published	true	2020-10-27 04:18:08	admin
Change - Normal - Assess	Global	Published	true	2020-10-06 05:37:05	admin
Change - Normal - Authorize	Global	Published	true	2020-10-06 05:38:35	admin
Change - Normal - Implement	Global	Published	true	2020-09-23 04:23:59	admin
Change - Standard	Global	Published	true	2020-09-23 05:09:01	admin

New

- Playbook
- Flow
- Subflow
- Trigger
- Action
- Decision table
- Help me decide

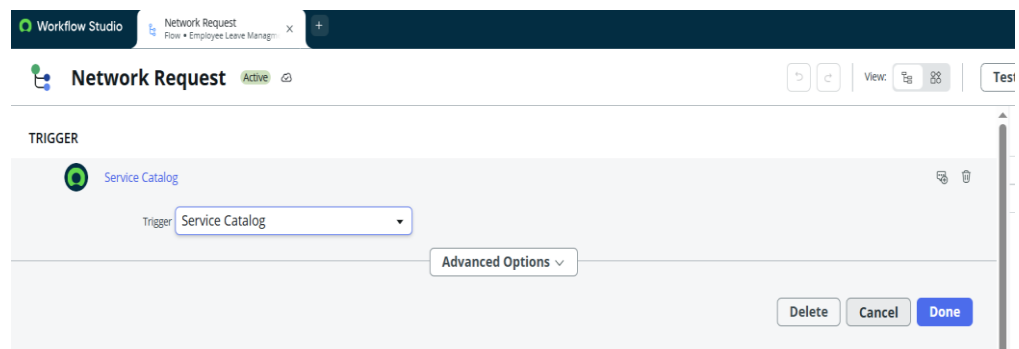
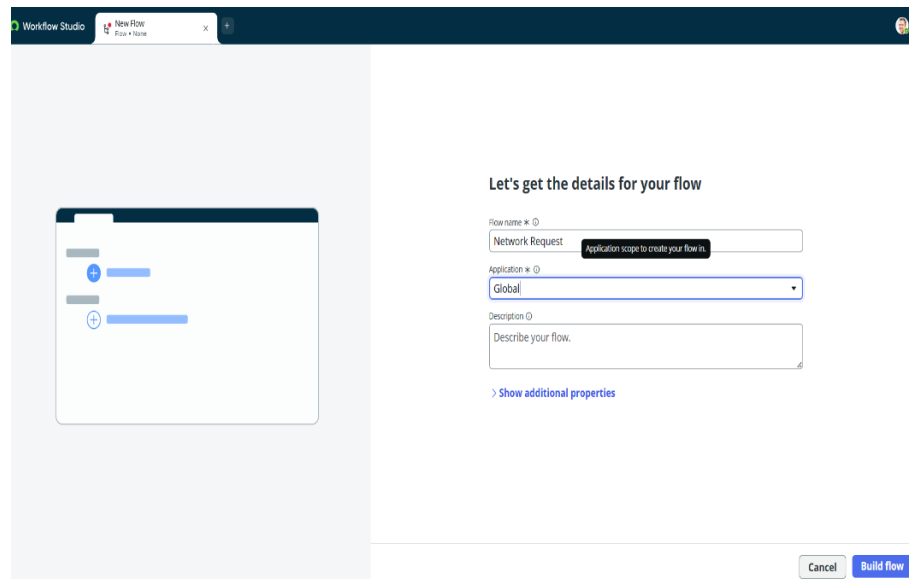
Pick up where you left off

- Network Request
Last updated: 4 d. ago by System Administrator...
- Leave Approval Flow
Last updated: 15 d. ago by System Administrator...
- Employee Leave Approval
Last updated: 15 d. ago by System Administrator...

Latest updates

- System Administrator modified Network Request 4 d. ago
- System Administrator modified Leave Approval Flow 15 d. ago
- System Administrator modified Employee Leave Approval 15 d. ago
- System Administrator modified Deployment Migration to ReleaseOps a year ago
- System Administrator modified Multi-factor Authentication a year ago

Resources

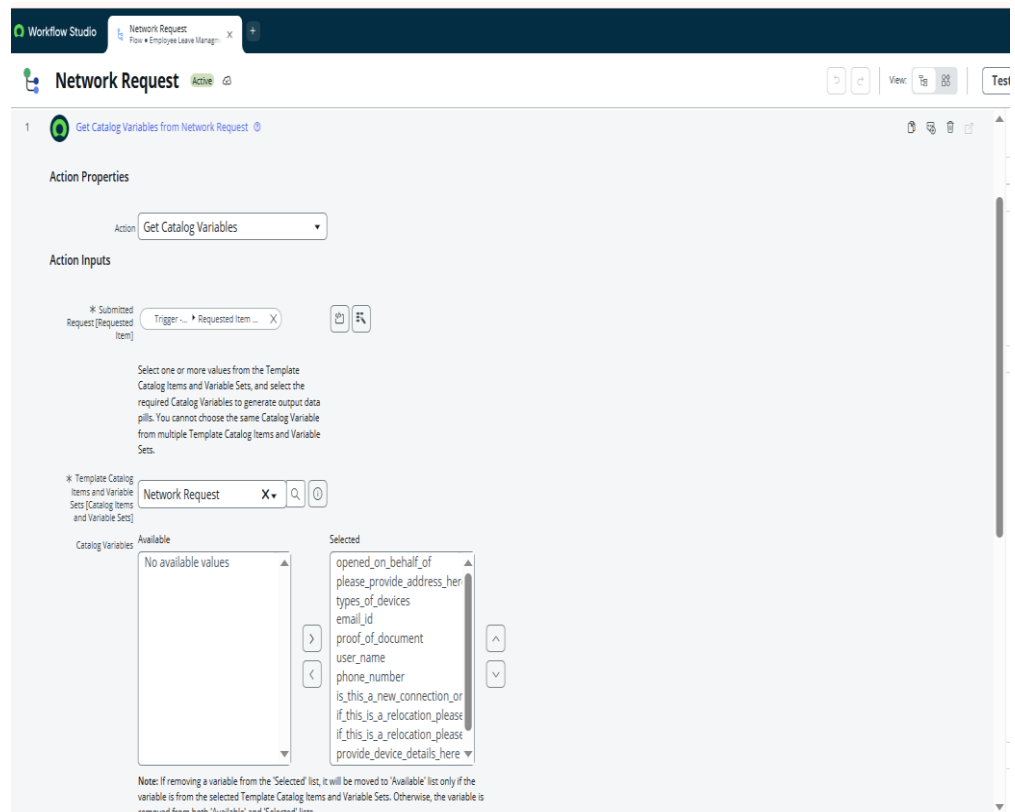


iii. Configuring Actions

Click on Actions button to configure new action

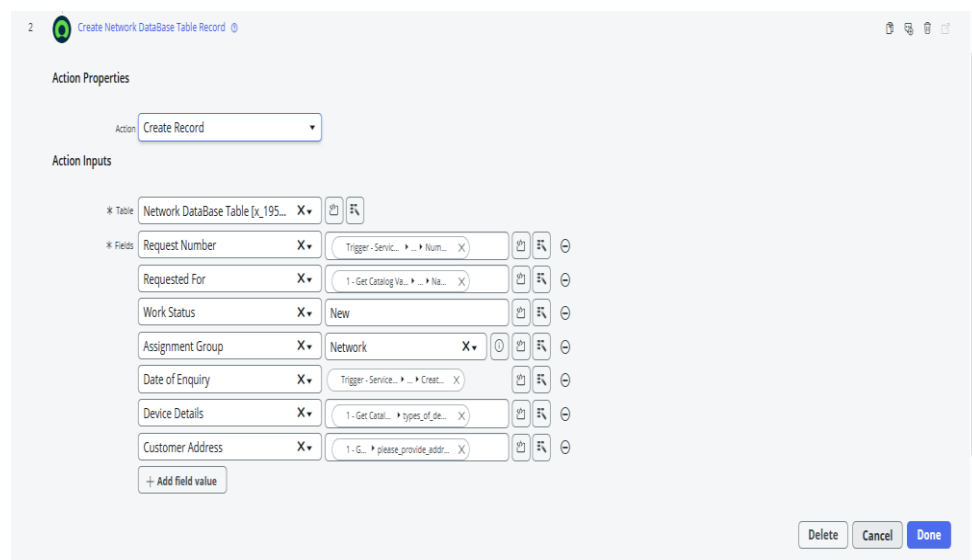
1. Get Catalog Variables

- Click on Action, search for Get Catalog Variables
- Select Get Catalog Variables
- Action Inputs >> Trigger >> service catalog >> Requested Item
- Template catalog items >> Select table >> Network Request
- Select the Required Variables and Move to the selected area.
- Click on done



2. Create Record

- Select action as Create Record
- Select table as Network Database
- Click on Add fields button to configure the fields
- Configure the Required fields as shown in the below picture
- Click on done



3. Ask for approvals

- Select action as Ask for Approval
- Select target record >> Create record>> network database table
- Provide Approval Reason>> Waiting for approval
- Configure approval rules>> Approve, reject, approve/reject
- Select approvals as Anyone approves, everyone approves etc.
- We can select approvals like static/dynamic as shown below
- Click on done

The screenshot shows the configuration interface for the 'Ask For Approval on Requested Item' action. The 'Action Properties' section shows the action name as 'Ask For Approval'. The 'Action Inputs' section includes fields for 'Record' (Trigger -> Requested Item), 'Table' (Requested Item [sc_req_item]), 'Approval Reason' (Waiting for Approval), 'Approval Field' (Approval), and 'Journal Field' (Approval history). The 'Rules' section shows a rule set with 'Approve' as the action, 'Anyone approves' as the condition, and 'Network X' as the target. The 'Due Date' is set to 'None'. The interface includes buttons for 'Delete', 'Cancel', and 'Done'.

4. Flow Logic

- Select action as flow logic and Select If condition
- Apply condition >> Ask for approvals state is Approved/Rejected as per requirement
- Click on done

The screenshot shows the configuration interface for the 'If' condition. The 'Condition Label' is 'If Request is Approved'. The 'Condition 1' is configured as 'Trigger -> Service -> Appro... X' is 'Approved'. The interface includes buttons for 'Delete', 'Cancel', and 'Done'.

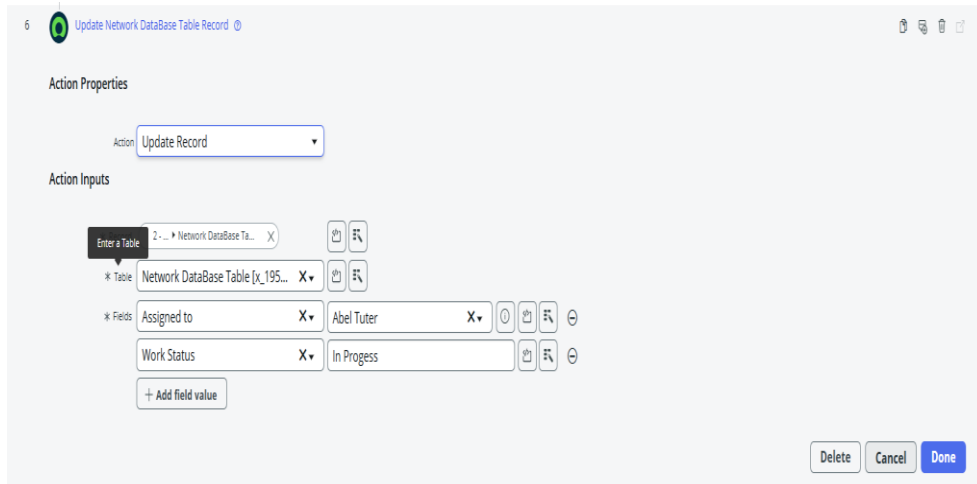
5. Send Email

- Select action as Send Email
- Select target record >> Create record>> network database table
- Table will be selected automatically
- Configure To, CC, BCC as per our requirements(select static/dynamic)
- Provide Subject & Body as shown in the below picture
- Click on done

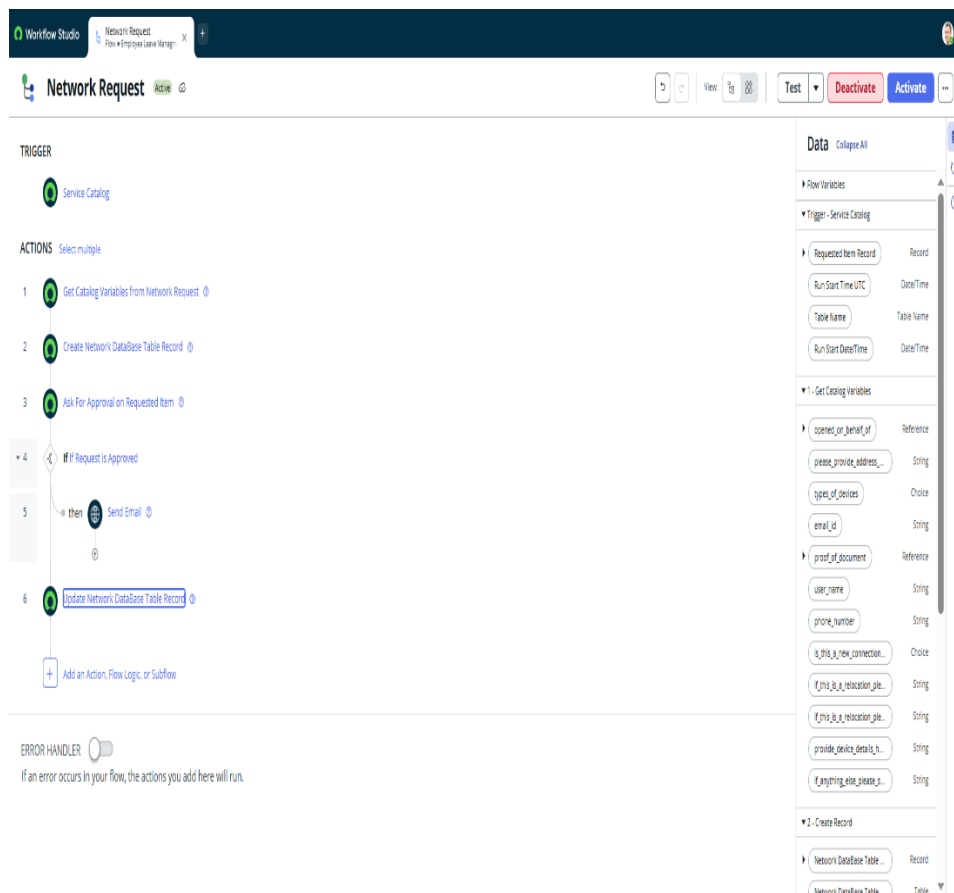
The screenshot shows the 'Send Email' configuration window. The 'Action' is set to 'Send Email'. Under 'Action Inputs', the 'Target Record' is '2 - Create Record -> Network Database Table', the 'Table' is 'Network Database Table (x_195...', and 'Include Watermark' is checked. The 'To' field is '1 - Get Catalog Va... -> Email...', 'CC' is '2 - Create Record -> Email...', 'BCC' is empty, and 'Subject' is 'Request had been created'. The 'Body' text area contains a greeting, a message about the request, an apology, and a signature for the 'Network Team'. The window has 'Delete', 'Cancel', and 'Done' buttons at the bottom right.

6.Update Record

- Select action as Update Record
- Select record as >> create record>> network database
- Table will be selected automatically
- Configure the fields as per requirement, as shown in below
- Click on done



iv. Flow Chart



V. Screenshots of Output

servicenow

AllFavoritesHistoryWorkspacesAdmin

Network Request

Application scope: Employee Leave Management
Update role: System Administrator | Employee Leave Manag...

Service Catalog > Standard Changes > Network Standard Changes > Network Request

Search catalog

Network request Management

User name

Opened on behalf of

Phone Number

Email Id

Proof of Document
Click to add...

Is this a New connection or Relocation?
☒ New
☐ Relocation
☐ None

If this is a relocation, Please provide...

If this is a relocation, Please provide...

Please provide address here

Types of Devices
☒ Laptop

Order this item

Quantity1

Delivery time2 Days

Order Now

Add to Cart

Shopping Cart

Empty

VI. Execution Details

Workflow Studio

Network Request

Network Request

EXECUTION DETAILS

Network Request

Test Run - Completed

Open flow

Open context record

Show Action Details

State

Start time

FLOW STATISTICS

Run as: System Administrator

Open flow logs

Completed

2024-08-15 13:32:19

1117ms

TRIGGER

Catalog Item Requested

ACTIONS

1

Get Catalog Variables from Network Request

Core Action

Completed

2024-08-15 13:32:19

18ms

2

Create Record

Core Action

Completed

2024-08-15 13:32:19

4ms

3

Ask For Approval

Core Action

Completed

2024-08-15 13:32:19

1064ms

4

If If Request is Approved

Flow Logic

Evaluated - True

2024-08-15 13:32:49

26ms

5

Send Email

Completed

2024-08-15 13:32:49

25ms

6

Update Record

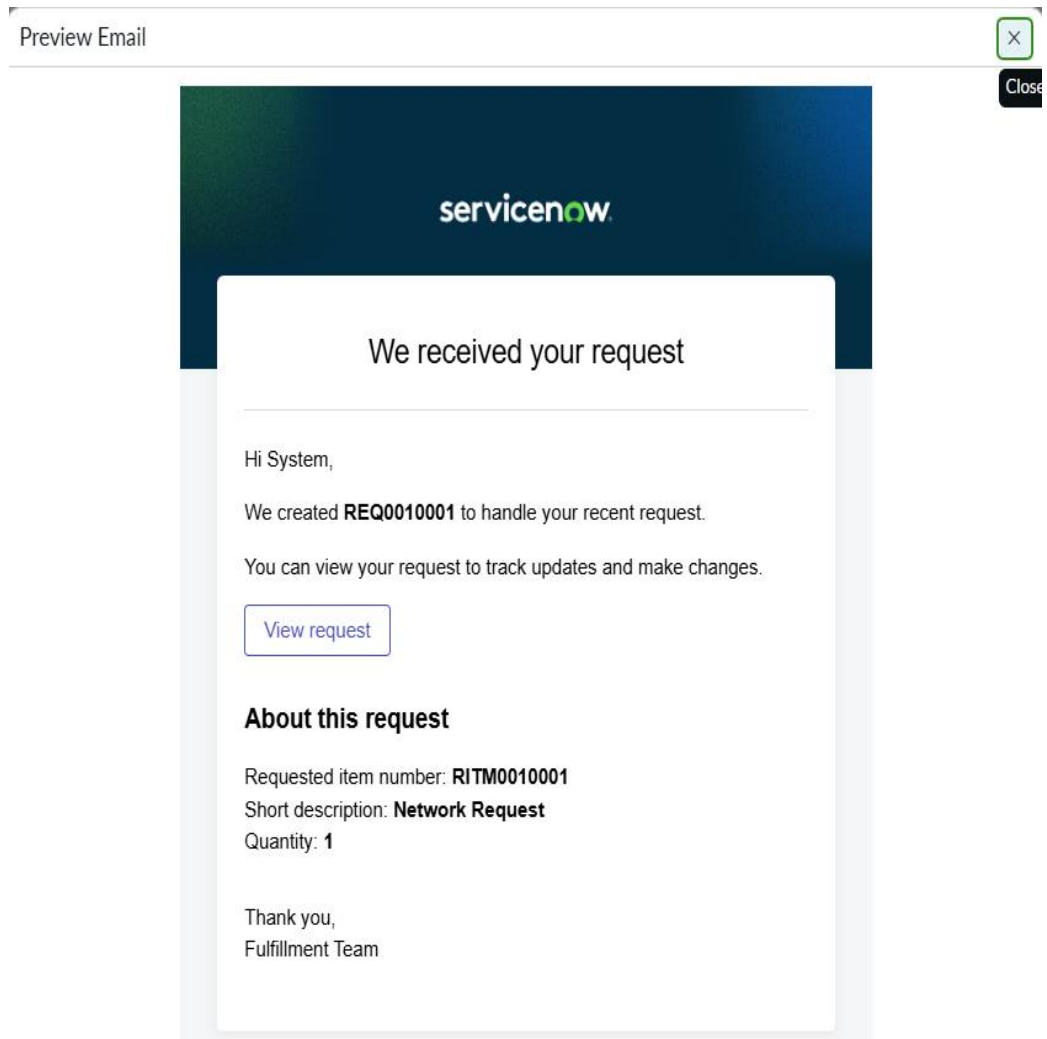
Core Action

Completed

2024-08-15 13:32:49

2ms

VII. EMAIL



VIII. Conclusion

The Automated Network Request Management in ServiceNow project provides a streamlined, reliable, and user-friendly solution for handling network-related service requests. By automating approvals, routing, and notifications, the system eliminates delays and errors associated with manual processes while offering transparency through real-time updates and reporting. This initiative not only improves SLA compliance and user satisfaction but also optimizes IT operations by reducing repetitive workload and enabling faster fulfillment. Ultimately, it supports the organization's goal of delivering efficient, high-quality, and responsive network services, empowering both end users and IT teams.